Masakhane Playbook

Democratizing machine translation for African languages

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Image Data

Multimodal Data

Speech Data

Text Data

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Data Storage and Release Infrastructure

Documentation and Reporting

Licensing and Compliance

Templates & Artifacts

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Synthetic Data Creation

LLM-Assisted Annotation

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Data Integrity and Contamination Control

Evaluation, Benchmarking, and Data Integrity

Model Building and Starter Kits

- 📄Maintenance and Post-Release Strategy
- 📄Release Checklist

Maintenance and Post-Release Strategy

- Dataset updates and versioning policy
- Deprecation strategy
- Community feedback loops

- Issue tracking

Release Checklist

- Data cleaned and validated
- Annotation quality verified
- Documentation completed
- Licensing defined
- Ethical review conducted
- Baselines and splits provided
- Public access ensured
- 📄Collaboration and Shared Tasks
- 📄Community Ecosystems

Collaboration and Shared Tasks

- Shared tasks and benchmarks
- Workshops and open challenges

Community Ecosystems

- Community initiatives (Masakhane, EthioNLP, HausaNLP)
- Academic and industry collaboration
- Contribution and contributor guidelines

1. Introduction

A comprehensive guide to dataset design, annotation, and task formulation for building reliable and responsible language AI systems.

Welcome to the dataset design and annotation playbook!

This playbook will help you plan and develop **training and evaluation datasets**, define **annotation schemas**, and design **AI tasks** across different languages, domains, and modalities. It provides guidance on dataset structuring, labeling strategies, and ethical considerations for language technologies.

Who is this playbook for?

This playbook is designed for:

- **Researchers** working on NLP dataset creation and evaluation
- **Annotation teams** developing labeled datasets
- **Project managers** overseeing data collection and annotation workflows
- **AI practitioners** designing and evaluating language models
- **Students and academics** studying dataset design and annotation
- **Multilingual communities** contributing to language resources

What will you learn?

By the end of this playbook, you will understand:

- How to define the **purpose and scope** of a dataset
- Differences between **training and evaluation datasets**
- Trade-offs between **scale and quality**
- How to design **label schemas and ontologies**

- Approaches for **multi-label, single-label, and structured outputs**
- How to handle **ambiguity, edge cases, and annotation boundaries**
- Best practices for **multilingual and cross-lingual dataset design**
- Ethical considerations, risks, and limitations in dataset creation

How to use this playbook

Each section of this playbook contains:

- **Clear explanations** of dataset design principles
- **Structured guidance** for task and schema definition
- **Examples and edge cases** to support annotation decisions
- **Practical recommendations** for dataset creation workflows
- **Ethical considerations** to guide responsible use

Getting Started

Ready to begin? Start with our foundational sections:

1. **Purpose of this Playbook** – Understand target users, scope, and intended use
2. **How to Use This Playbook** – Learn how to navigate chapters and contribute
3. **Dataset Types and Design Goals** – Explore dataset categories and trade-offs
4. **Task and Schema Definition** – Define tasks, labels, and annotation structures
5. **Glossary and Terminology** – Learn key concepts and definitions

Purpose of this playbook

- Target users and communities
- Languages, domains, and modalities covered
- Intended use and risks

Dataset Types and Design Goals

- Training vs evaluation datasets
- General-purpose vs domain-specific datasets
- Scale vs quality trade-offs
- Monolingual, multilingual, cross-lingual setups

Task and Schema Definition

- Task formulation (classification, generation, alignment, retrieval)
- Label schema and ontology design
- Multi-label vs single-label vs structured outputs
- Ambiguity, edge cases, and annotation boundaries

Glossary and Terminology

A reference section providing clear definitions of key terms used throughout the playbook.

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Cost and Resource Planning

Learn how to effectively plan the resources required for dataset creation, including budgeting, timelines, an...

Data Cleaning and Preprocessing

Learn how to prepare raw data for use in language AI systems by improving quality, consistency, and usabil...

Data Provenance and Traceability

Learn how to track the origin, history, and transformations of your data to ensure transparency, reproducibil...

Ethics, Bias, and Governance

Learn how to ensure responsible dataset creation by addressing bias, protecting privacy, and maintaining tr...

Cost and Resource Planning

Learn how to effectively plan the resources required for dataset creation, including budgeting, timelines, and scaling strategies.

Why Planning Matters

Dataset creation can be resource-intensive. Proper planning helps ensure efficient use of time, budget, and human effort while maintaining data quality.

Key Components of Planning

Budgeting Annotation Costs

- **Annotation cost estimation** – Calculate cost per sample or per task
- **Workforce planning** – Consider expert vs crowd annotators
- **Tooling costs** – Include platforms, storage, and infrastructure
- **Quality control costs** – Account for validation and review processes

Time Estimation

- **Task complexity** – More complex tasks require more time per annotation
- **Annotator speed** – Estimate based on pilot studies or benchmarks
- **Project phases** – Include setup, training, annotation, and validation
- **Buffer time** – Plan for delays and iterations

Scaling Strategy

- **Incremental scaling** – Start small and expand gradually
- **Automation support** – Use tools to speed up preprocessing and validation
- **Parallel workflows** – Distribute tasks across multiple annotators
- **Quality vs scale balance** – Maintain data quality while increasing size

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Data Cleaning and Preprocessing

Learn how to prepare raw data for use in language AI systems by improving quality, consistency, and usability.

Why Data Cleaning Matters

Raw data often contains noise, inconsistencies, and errors. Proper cleaning and preprocessing ensure that datasets are reliable, accurate, and suitable for downstream tasks such as training and evaluation.

Key Steps in Data Cleaning

Deduplication, Normalization, and Filtering

- **Deduplication** – Remove duplicate entries to avoid bias and overrepresentation
- **Normalization** – Standardize text (e.g., casing, punctuation, encoding)
- **Filtering** – Remove irrelevant, low-quality, or out-of-scope data

Language Detection and Formatting

- **Language detection** – Identify and verify the language of each data instance
- **Formatting** – Ensure consistent structure (e.g., JSON, CSV, text fields)
- **Encoding consistency** – Maintain uniform character encoding (e.g., UTF-8)

Noise and Toxicity Handling

- **Noise removal** – Clean unwanted artifacts such as HTML tags, emojis (if not needed), or corrupted text
- **Toxicity handling** – Detect and manage harmful, offensive, or unsafe content depending on project goals

Missing and Corrupted Data Handling

- **Missing data** – Identify incomplete entries and decide whether to fill, ignore, or remove them
- **Corrupted data** – Detect broken or unreadable content and clean or discard it
- **Validation checks** – Ensure data integrity after preprocessing

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Data Provenance and Traceability

Learn how to track the origin, history, and transformations of your data to ensure transparency, reproducibility, and accountability.

Why Provenance Matters

Understanding where data comes from and how it has been processed is essential for building trustworthy datasets. Provenance supports reproducibility, enables auditing, and helps identify potential issues in data quality and bias.

Key Components of Provenance

Source Tracking

- **URLs and references** – Record links or original sources of the data
- **Contributors** – Track who collected, created, or provided the data
- **Collection context** – Document when, where, and how the data was obtained

Data Lineage

- **Data evolution** – Track how data changes over time
- **Versioning** – Maintain different versions of datasets
- **Pipeline tracking** – Document each stage of data processing

Transformation Logs

- **Preprocessing steps** – Record cleaning, normalization, and filtering operations
- **Annotation processes** – Track labeling methods and guidelines used
- **Modifications** – Log any changes made to the data after collection

- **Audit trails** – Maintain records for reproducibility and verification

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Ethics, Bias, and Governance

Learn how to ensure responsible dataset creation by addressing bias, protecting privacy, and maintaining transparency throughout the data lifecycle.

Why Ethics and Governance Matter

Datasets directly influence the behavior of language AI systems. Poor handling of bias, privacy, or sensitive content can lead to harmful outcomes. Ethical practices and governance frameworks help ensure fairness, accountability, and trust.

Key Components of Ethical Data Practices

Bias Identification and Mitigation

- **Bias detection** – Identify imbalances or skewed representations in data
- **Source bias** – Assess biases introduced by data sources
- **Annotation bias** – Monitor inconsistencies across annotators
- **Mitigation strategies** – Apply re-sampling, re-weighting, or guideline refinement

PII Detection and Removal

- **Personal data identification** – Detect names, addresses, contact details, and identifiers
- **Automated detection tools** – Use models or rules to flag sensitive information
- **Manual review** – Validate automated detection with human checks
- **Data removal or masking** – حذف or obfuscate personal identifiers

Anonymization Strategies

- **De-identification** – Remove or replace identifiable information
- **Pseudonymization** – Substitute identifiers with artificial labels
- **Aggregation** – Present data in grouped form to prevent re-identification

- **Risk assessment** – Evaluate re-identification risks after anonymization

Sensitive Attribute and Content Handling

- **Sensitive attributes** – Gender, ethnicity, religion, health, or political views
- **Content moderation** – Handle harmful, offensive, or explicit content carefully
- **Access control** – Restrict sensitive data to authorized users
- **Use-case alignment** – Decide inclusion based on task requirements

Fair Representation

- **Inclusive sampling** – Ensure diverse representation across groups
- **Balanced datasets** – Avoid over- or under-representation
- **Context awareness** – Consider cultural and linguistic diversity
- **Evaluation fairness** – Test models across different subgroups

Risk Documentation and Transparency

- **Risk identification** – Document potential harms and limitations
- **Datasheets and documentation** – Provide clear dataset descriptions
- **Transparency practices** – Share collection, processing, and annotation details
- **Governance policies** – Define rules for dataset usage and distribution

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Annotation Task Design and Human Factors

Task complexity estimation

Inclusive and Bias-Aware Annotation

Learn how to design annotation processes that are inclusive, fair, and aware of potential biases introduced ...

Training and Guidelines

Learn how to prepare annotators through clear instructions, structured training, and iterative refinement of ...

Workflow and Adjudication

Multi-annotator setup

Annotation Task Design and Human Factors

Task complexity estimation

Annotator fatigue

UI/UX considerations

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Inclusive and Bias-Aware Annotation

Learn how to design annotation processes that are inclusive, fair, and aware of potential biases introduced by annotators and data.

Why Inclusion and Bias Awareness Matter

Annotation is a human-driven process, and annotator backgrounds can influence labeling decisions. Ensuring diversity and awareness helps reduce bias and improves the quality and fairness of datasets.

Key Components

Gender, Age, and Cultural Diversity

- Include annotators from diverse **gender**, **age groups**, and **cultural backgrounds**
- Ensure representation across different **dialects and communities**
- Avoid over-reliance on a single demographic group
- Consider cultural context when interpreting data

Recording Annotator Personas

- Document annotator characteristics where appropriate and ethical
- Capture information such as **language background**, **region**, or **expertise**
- Use anonymized metadata to analyze potential annotation biases
- Ensure privacy and consent when collecting annotator information

Bias Awareness Training

- Train annotators to recognize **personal and cultural biases**

- Provide examples of biased vs unbiased annotations
- Encourage consistent application of guidelines
- Reinforce neutrality and objectivity in labeling

Community Participation

- Engage local communities in the annotation process
- Incorporate native speaker knowledge and cultural insights
- Promote participatory and inclusive dataset creation
- Respect community norms and values throughout the process

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Training and Guidelines

Learn how to prepare annotators through clear instructions, structured training, and iterative refinement of annotation tasks.

Annotation Guidelines with Examples

Clear and consistent guidelines are the foundation of reliable annotation.

- Define each label or category in simple and unambiguous terms
- Provide **positive examples** for each label
- Provide **negative examples** to clarify boundaries
- Include **edge cases** to handle ambiguity
- Specify how to treat uncertain or mixed cases
- Use consistent formatting and terminology throughout the guideline

Training and Calibration Rounds

Training ensures annotators understand and apply guidelines consistently.

- Conduct initial training sessions before annotation begins
- Use calibration tasks to align annotator understanding
- Compare annotations across multiple annotators for the same samples
- Provide structured feedback to resolve misunderstandings
- Repeat calibration until acceptable agreement is reached

Pilot Annotation and Iteration

Pilot annotation helps test and refine the annotation design before scaling.

- Start with a small subset of data
- Identify unclear instructions or confusing labels

- Measure annotation consistency and difficulty
- Collect annotator feedback on task clarity
- Iteratively refine guidelines, labels, and workflow

Minimum Viable Dataset per Task

A minimum viable dataset ensures the task design is valid before full-scale annotation.

- Create a small but representative dataset for each task
- Validate label schema coverage and clarity
- Test annotation workflow and tool usability
- Check feasibility of large-scale annotation
- Use results to decide whether to scale or redesign the task

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Workflow and Adjudication

Multi-annotator setup

Task assignment and redundancy

Disagreement resolution and expert adjudication

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Data Quality Management

Class imbalance handling

Data Quality Assurance

Inter-Annotator Agreement (Kappa, Alpha)

Data Quality Management

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Outlier and noise detection

Error analysis pipelines

Dataset versioning and updates

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Data Quality Assurance

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Gold data and control checks

Auditing and spot-checking

Feedback loops

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Image Data

- Image Classification and Recognition – (object classification, scene recognition)

Multimodal Data

- Vision-Language Tasks – (image-text retrieval, captioning, VQA)

Speech Data

- ASR (Automatic Speech Recognition) – (transcription, multilingual ASR, code-switching)

Text Data

- Text Classification – (sentiment, emotion, hate speech, topic, intent detection)

Image Data

- **Image Classification and Recognition** – (object classification, scene recognition)
- **Object Detection and Segmentation** – (bounding boxes, instance/semantic segmentation)
- **Image Captioning and Generation** – (captioning, image-to-text generation)
- **Vision-Language Tasks** – (Visual Question Answering (VQA), referring expressions)
- **Image-to-Image Tasks** – (style transfer, super-resolution, restoration)
- **Document Understanding** – (OCR, layout analysis, form understanding)

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Multimodal Data

- **Vision-Language Tasks** – (image-text retrieval, captioning, VQA)
- **Audio-Text Tasks** – (speech translation, audio captioning)
- **Cross-Modal Retrieval and Alignment** – (text-to-image, image-to-text search)
- **Multimodal Generation** – (text-to-image, text-to-video, image-conditioned text generation)
- **Reasoning and Instruction Tasks** – (multimodal QA, instruction following)

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Speech Data

- **ASR (Automatic Speech Recognition)** – (transcription, multilingual ASR, code-switching)
- **TTS (Text-to-Speech)** – (single-speaker, multi-speaker, expressive TTS)
- **Speech-to-Speech Translation (STS)** – (direct speech translation across languages)
- **Audio Understanding** – (audio classification, sound event detection)
- **Speech emotion recognition**
- **Speaker diarization**

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Text Data

- **Text Classification** – (sentiment, emotion, hate speech, topic, intent detection)
- **Sequence Labeling** – (NER, POS tagging, chunking, slot filling, keyphrase extraction)
- **Sequence-to-Sequence** – (machine translation, summarization, paraphrasing, simplification)
- **Question Answering and Reasoning** – (extractive QA, generative QA, reading comprehension)
- **Retrieval and Ranking** – (document retrieval, semantic search, reranking)
- **Dialogue and Generation** – (chatbots, instruction following, story generation)
- **Structured Prediction and Parsing** – (dependency parsing, constituency parsing, semantic parsing)

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Data Storage and Release Infrastructure

- Repository hosting (Hugging Face, GitHub, institutional)

Documentation and Reporting

- Datasheets for datasets

Licensing and Compliance

- Licensing types and attribution

Templates & Artifacts

- Annotation guideline template

Tooling and Infrastructure

- Annotation tools (usability, scalability, cost)

Data Storage and Release Infrastructure

- Repository hosting (Hugging Face, GitHub, institutional)
- File formats and metadata standards
- Versioning and changelogs

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Documentation and Reporting

- Datasheets for datasets
- Standard reporting and reproducibility
- Failure cases and limitations
- Transparency in dataset creation

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Licensing and Compliance

- Licensing types and attribution
- Usage restrictions
- Legal and ethical compliance

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Templates & Artifacts

- Annotation guideline template
- Datasheet template
- Data collection plan template
- Budget estimator

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Tooling and Infrastructure

- Annotation tools (usability, scalability, cost)
- Data pipelines and automation
- Deployment (cloud vs local)
- Security and access control

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Synthetic Data Creation

- Data augmentation - (paraphrasing, back-translation)

LLM-Assisted Annotation

- LLM-assisted annotation - (human-in-the-loop)

Synthetic Data Creation

- **Data augmentation** - (paraphrasing, back-translation)
- **Fully synthetic generation** - (LLMs, simulation)
- **Scenario-based data generation**
- **Validation against real-world distributions**
- **Evaluation of synthetic data quality**

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LLM-Assisted Annotation

- **LLM-assisted annotation** - (human-in-the-loop)
- **LLM and a human annotator agreement**
- **Prompt design and output validation**
- **Bias, hallucination, and consistency control**
- **When NOT to use LLMs**
- **Cost vs quality comparison** (LLM vs human)

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Data Integrity and Contamination Control

- Preventing train-test leakage

Evaluation, Benchmarking, and Data Integrity

- Evaluation Metrics by task

Data Integrity and Contamination Control

- Preventing train-test leakage
- Overlap with existing benchmarks
- LLM contamination (training data exposure)

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Evaluation, Benchmarking, and Data Integrity

- Evaluation Metrics by task
- Train/dev/test splits
- Cross-lingual and domain generalization
- Bias and robustness evaluation
- Bias evaluation metrics

Model Building and Starter Kits

- Baseline models for each modality/task
- Training and evaluation scripts
- Reproducibility guidelines
- Benchmark leaderboards
- Benchmark positioning

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Maintenance and Post-Release Strategy

Dataset updates and versioning policy

Release Checklist

Data cleaned and validated

Maintenance and Post-Release Strategy

Dataset updates and versioning policy

Deprecation strategy

Community feedback loops

Issue tracking

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Release Checklist

Data cleaned and validated

Annotation quality verified

Documentation completed

Licensing defined

Ethical review conducted

Baselines and splits provided

Public access ensured

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Collaboration and Shared Tasks

Shared tasks and benchmarks

Community Ecosystems

Community initiatives (Masakhane, EthioNLP, HausaNLP)

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